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Where the Blue Angels roar comes from

Requested by the Falcone family, loyal readers



Blue Angels over Lake Washington, August 1, 2025. The formation is visible; most of the sound-producing exhaust flow is not. U.S. Navy photo by MC1 Shane Bryan, 250801-N-AD724-1133 (public domain). Photo source.

A family question with a surprisingly good twist

The jets cross the sky, the roar swells, and the question practically asks itself: how much of that sound is the engines, and how much is the airplane pushing through the air? For a powerful fighter flyby, the exhaust is generally the main contributor to the spectacular roar. But much of that engine noise is itself made by moving air. The useful distinction is **propulsion noise versus airframe noise**.

The Blue Angels fly F/A-18 Super Hornets [1]. Their engines accelerate gas, while the wings, fuselage and control surfaces redirect the surrounding airflow. Both can produce sound. Today we will separate those contributions, explain the crackle and changing pitch, and see why an honest answer cannot be a fixed percentage for every pass.

The roar continues beyond the nozzle

Picture a fast river entering slower water. The boundary curls into swirls that stretch, merge and break apart. Behind a fighter, the fast exhaust does something similar in three dimensions. Its unsteady turbulence produces pressure disturbances that radiate outward as sound. The loudspeaker is effectively a region of moving gas extending behind the engine, rather than a single opening in the metal.

This is **turbulent mixing noise**. It spans a broad range of frequencies, giving a roar rather than a musical note. Large flow structures tend to contribute at lower frequencies; smaller, faster changes contribute higher-frequency content. High-speed jet research identifies powerful large-scale turbulent structures and strong radiation toward the downstream, or aft, direction [2]. That helps explain why the departure side of a flyby can be especially forceful.

Supersonic exhaust adds another ingredient. If the pressure leaving the nozzle does not match the surrounding air, the plume can contain a repeating pattern of compression and expansion, including shock cells. Turbulence interacting with that pattern creates **broadband shock-associated noise**. Under some conditions a feedback loop produces a more tonal sound called screech. These are distinct jet-noise mechanisms, and none should be diagnosed from a single adjective such as “harsh” [3].

The machinery also has a voice

Inside the engine, rotating fan and compressor blades and their interactions with stationary vanes produce tones and broadband sound. Turbines and unsteady combustion contribute too [4]. A whine may make these rotating components easier to notice when the exhaust roar is less dominant. Calling all engine sound “fuel exploding” misses both the moving-blade sources and the turbulent plume outside the engine.

Afterburners inject and burn extra fuel downstream of the turbine to add energy and thrust [5]. They change the exhaust temperature, flow and nozzle operating condition, often making a fighter markedly louder. But afterburner is not an on/off switch for all jet noise: a fighter at high power without afterburner can still be extremely loud. Nor is there one universal decibel penalty that applies at every angle and operating point.

Three places to listen for, conceptually

Inside the engine	Behind the nozzle	Around the aircraft
Rotating blades, vane interactions and combustion.	Turbulent exhaust mixing; shock-related noise when conditions support it.	Unsteady airflow interacting with wings, edges, gaps and landing gear.
Propulsion noise	Propulsion noise, made aerodynamically	Airframe noise

Original source guide. Categories explain mechanisms; their widths do not represent equal sound contributions.

Even a silent engine would leave a whoosh

An airplane moving through air would make sound even if its propulsion system were acoustically silent. The thin layer of air along the surface becomes turbulent. When its fluctuating pressure field encounters an edge, some of that disturbance radiates as sound. Exposed landing gear creates wakes around wheels, struts and braces. Flap edges, gaps and open cavities add more unsteady flow and possible resonances [6].

This is more specific than “air rubbing on metal.” The important ingredients are changing pressure and force, and how the flow interacts with geometry. A wing that generates lift does not automatically emit a roar proportional to that lift. Vortices and separated flow can create unsteady loading, but a dramatic-looking vortex is not a sound-level meter.

Configuration changes the balance

A fast, clean fighter with gear retracted and substantial engine power has a very different sound budget from an aircraft approaching a runway with gear and flaps deployed and engines throttled back. In the latter case, airframe noise becomes more important relative to propulsion. NASA demonstrated this distinction by flying a Gulfstream research aircraft at ground-idle thrust over a microphone array to evaluate quieter flaps, gear fairings and cavity treatments [6]. Those tests establish real airframe contributions; they do not provide a percentage breakdown for a Blue Angels maneuver.

The two categories can also interact. An airframe can reflect or shield engine sound, and an exhaust plume passing near a surface can generate additional interaction noise [7]. Engineers therefore distinguish the original source from the path by which it reaches a listener. Moving a surface may change both.

So, was that a sonic boom?

The Blue Angels state that their air-show demonstrations are not authorized to exceed the speed of sound [1]. A startling high-speed pass is therefore not, by itself, evidence of a sonic boom. The aircraft can remain subsonic while its exhaust is supersonic. NASA recorded both mixing noise and broadband shock noise during subsonic F-18 and F-16XL flyovers [8].

The ripping or popping quality called **jet crackle** is another distinction. Research links it to very steep pressure changes and acoustic shocks; intense sound can steepen nonlinearly as it travels [9]. These sharp features in jet noise differ from the aircraft-scale shock system responsible for a supersonic flyover boom. “I heard a crack” is an observation, not enough evidence to identify the mechanism.

Two sources do not split the decibels

Suppose, purely for illustration, that propulsion alone would produce 100 dB at a listener and airframe noise alone would produce 90 dB. Assume the same frequency band, weighting and averaging period, and sources whose pressure fluctuations are uncorrelated. These are invented teaching values, not Blue Angels measurements. The combined level is

$$L_{\text{total}} = 10\log_{10}(10^{L_{\text{engine}}/10} + 10^{L_{\text{airframe}}/10})$$

$$L_{\text{total}} = 10\log_{10}(10^{10} + 10^{9}) \approx 100.41 \text{ dB}$$

The quieter source adds only 0.41 dB. In this example, propulsion supplies about 91% of the summed mean-square pressure and the airframe about 9%. Those are energy-related shares at that receiver, not percentages of perceived loudness. Removing the airframe contribution entirely would barely change the total level.

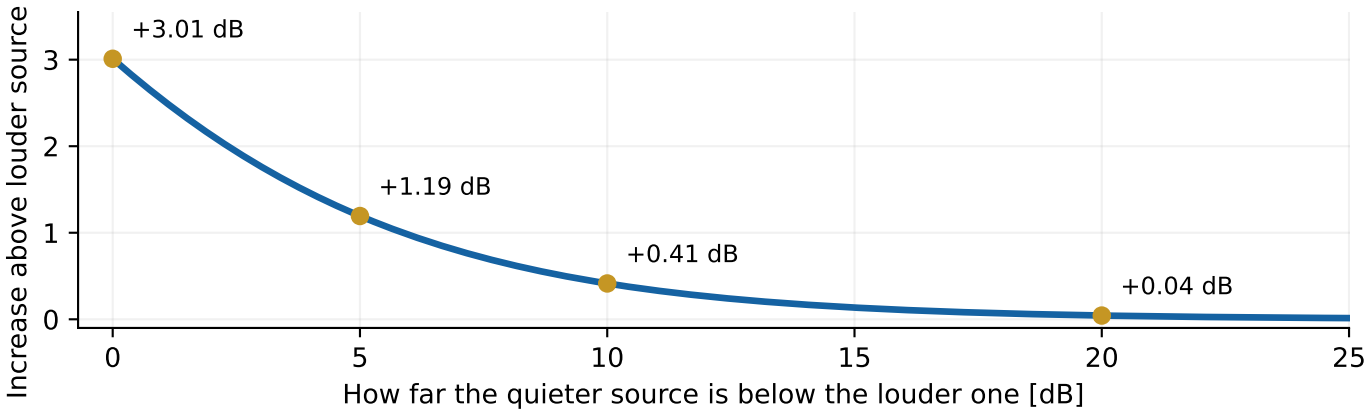


Figure 1. Original calculation for two uncorrelated contributions. Equal levels add 3.01 dB; a source 20 dB below the dominant one adds only 0.04 dB. This is an addition rule, not a measured fighter spectrum.

Now lower the engine contribution until it equals the airframe contribution. Each supplies half the mean-square pressure, and removing either reduces the total by about 3 dB. The physical airplane need not change for its apparent source balance to change: throttle, speed, configuration, frequency and viewing angle all matter.

A defensible Blue Angels percentage would require measurements for the specific maneuver, observer location and acoustic metric. Microphone arrays, synchronized flight and engine data, and carefully controlled comparison runs help separate sources. One phone recording usually cannot: automatic gain control, clipping and microphone directionality can change the apparent balance. Even a calibrated total sound level does not identify where the sound originated.

Why the sound seems to chase the jet

Light reaches you effectively immediately; sound needs time. In still air near room temperature, a 500 m sound path takes roughly 1.5 seconds. By the time that sound arrives, a fast aircraft has moved a long way. You are hearing an earlier position and operating condition, not necessarily the airplane you see at that instant.

Motion also changes pitch through the Doppler effect: approaching wavefronts arrive more closely spaced; receding ones arrive farther apart. At the same time, distance, engine power and the direction in which the plume radiates most strongly change the level. A roar that peaks after the visual pass is consistent with sound-travel delay and aft-directed radiation; it does not prove that the pilot suddenly added throttle.

For the Falcone family, the useful listening question is: which ingredient changed? A deeper departure roar suggests the exhaust contribution and viewing angle. A whine points toward tonal machinery or another tonal source. A change with gear or flap deployment invites an airframe explanation. These are clues, not source identifications from sound alone.

Three things to carry to the next air show

1. The powerful fighter roar is generally led by propulsion, especially turbulent exhaust mixing.
2. Airframe noise is real; its relative importance changes with speed, configuration, power and observer position.
3. Crackle, Doppler shift and delayed sound arrival can make a subsonic flyby dramatic without an aircraft sonic boom.

A question for the readers: If the jet were sitting still during an engine test, could it still roar?
Yes. The exhaust would still rush through surrounding air. That is why “air moving” and “airplane moving” are different parts of this story.

Sources and further reading

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